

ENTRY 075 Wiring the Demo to the Live API — September 5, 2026 — Status: COMPLETE

A Progressive Upgrade, Not a Hard Cutover

The Demo Works Identically Today and Upgrades Automatically Once Deployed

What Changed

quantumbridge_live_demo.html now tries a live request to the Entry 074 API first (4-second timeout via AbortController), and only falls back to the precomputed 24,003-row table if the API is unreachable or a base URL hasn't been configured. Every result is now labeled with an explicit badge -- green "live prediction" or grey "cached / precomputed" -- so a viewer always knows which path served their answer, rather than the two being silently indistinguishable. Because API_BASE_URL defaults to an empty string, the demo's behavior is completely unchanged until someone deploys the API and sets that one constant -- this is a progressive upgrade wired in ahead of time, not a dependency that had to ship together with a live deployment.

CORS

The API (Entry 074) added CORSMiddleware with a permissive allow_origins=["*"] policy, verified locally with a real preflight OPTIONS request carrying the GitHub Pages origin header -- confirmed the server responds with the correct access-control-allow-origin and access-control-allow-methods headers. This is deliberately permissive since /predict is a public, read-only, no-auth endpoint with no user data; it should be tightened before adding anything that isn't purely read-only.

What Ground Truth Display Still Relies On

Live requests don't run a fresh Aer simulation per pair (too slow for interactive use), so the "Simulated hardware result" field still comes from the local precomputed table when available, merged in alongside the live v4.1/GNN predictions -- a live prediction can still show real ground truth if that specific pair happened to be simulated during training or evaluation, it just won't trigger a brand-new simulation on demand.

Entry 075 — Conclusion

The frontend is ready for a live backend the moment one exists at a real URL -- deployment itself (choosing and configuring a host) remains the user's step, documented in api/README.md's new "Wiring up the live demo" section. Nothing about the current, already-deployed demo changes until that URL is set.

Research Log — Index Addendum (Entry 075)

Files: quantumbridge_live_demo.html (API_BASE_URL, fetchLive(), badge system), api/main.py (CORS middleware), api/README.md (wiring instructions).